

Class19

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Background

Pertussis is a bacterial lung infection also known as whooping cough. Let's begin by examining CDCD reported case numbers in the US.

Q1. With the help of the R “addin” package datapasta assign the CDC pertussis case number data to a data frame called cdc and use ggplot to make a plot of cases numbers over time.

```
cdc <- data.frame(  
  year = c(1922L,1923L,1924L,1925L,  
           1926L,1927L,1928L,1929L,1930L,1931L,  
           1932L,1933L,1934L,1935L,1936L,  
           1937L,1938L,1939L,1940L,1941L,1942L,  
           1943L,1944L,1945L,1946L,1947L,  
           1948L,1949L,1950L,1951L,1952L,  
           1953L,1954L,1955L,1956L,1957L,1958L,  
           1959L,1960L,1961L,1962L,1963L,  
           1964L,1965L,1966L,1967L,1968L,1969L,  
           1970L,1971L,1972L,1973L,1974L,  
           1975L,1976L,1977L,1978L,1979L,1980L,  
           1981L,1982L,1983L,1984L,1985L,  
           1986L,1987L,1988L,1989L,1990L,
```

```

1991L, 1992L, 1993L, 1994L, 1995L, 1996L,
1997L, 1998L, 1999L, 2000L, 2001L,
2002L, 2003L, 2004L, 2005L, 2006L, 2007L,
2008L, 2009L, 2010L, 2011L, 2012L,
2013L, 2014L, 2015L, 2016L, 2017L, 2018L,
2019L, 2020L, 2021L, 2022L, 2023L, 2024L),
cases = c(107473, 164191, 165418, 152003,
202210, 181411, 161799, 197371,
166914, 172559, 215343, 179135, 265269,
180518, 147237, 214652, 227319, 103188,
183866, 222202, 191383, 191890, 109873,
133792, 109860, 156517, 74715, 69479,
120718, 68687, 45030, 37129, 60886,
62786, 31732, 28295, 32148, 40005,
14809, 11468, 17749, 17135, 13005, 6799,
7717, 9718, 4810, 3285, 4249, 3036,
3287, 1759, 2402, 1738, 1010, 2177, 2063,
1623, 1730, 1248, 1895, 2463, 2276,
3589, 4195, 2823, 3450, 4157, 4570,
2719, 4083, 6586, 4617, 5137, 7796, 6564,
7405, 7298, 7867, 7580, 9771, 11647,
25827, 25616, 15632, 10454, 13278,
16858, 27550, 18719, 48277, 28639, 32971,
20762, 17972, 18975, 15609, 18617,
6124, 2116, 3044, 7063, 22538)
)
head (cdc)

```

```

  year cases
1 1922 107473
2 1923 164191
3 1924 165418
4 1925 152003
5 1926 202210
6 1927 181411

```

We can check whether we can plot cases per year for pertussis in the US

```

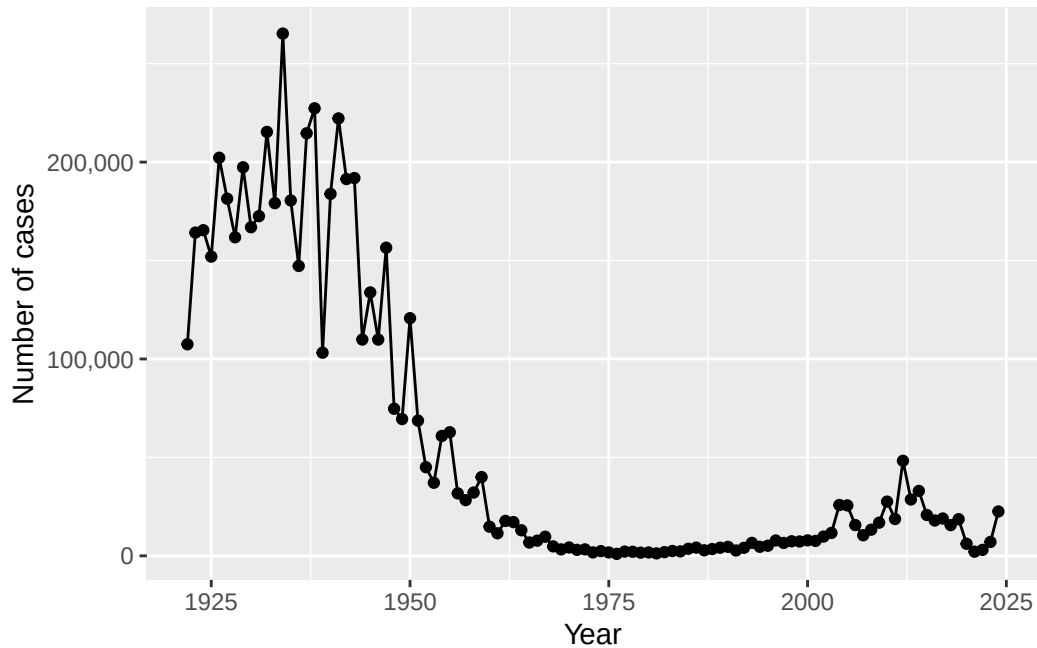
library(scales)
library(ggplot2)
ggplot(cdc) +

```

```

aes(year, cases) +
geom_point() +
geom_line() +
scale_y_continuous(labels = comma) +
labs(x = "Year",
      y = "Number of cases")

```



Q2. Using the ggplot `geom_vline()` function add lines to your previous plot for the 1946 introduction of the wP vaccine and the 1996 switch to aP vaccine (see example in the hint below). What do you notice?

Add some major milestone timepoints to our plot.

```

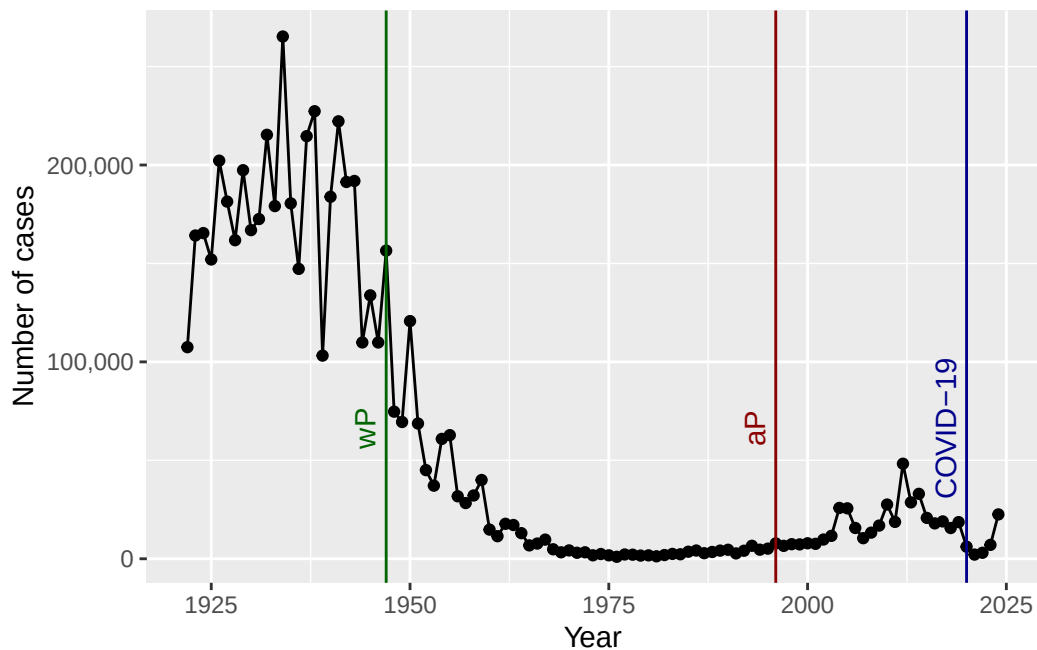
ggplot(cdc) +
  aes(year, cases) +
  geom_point() +
  geom_line() +
  scale_y_continuous(labels = comma) +
  labs(x = "Year",
        y = "Number of cases") +

  # vertical lines
  geom_vline(xintercept = 1947, color = "darkgreen") +

```

```
geom_vline(xintercept = 1996, color = "darkred") +
geom_vline(xintercept = 2020, color = "darkblue") +

# lower labels (25% up the plot)
annotate("text", x = 1947, y = max(cdc$cases)*0.25,
        label = "wP", color = "darkgreen", angle = 90, vjust = -0.5) +
annotate("text", x = 1996, y = max(cdc$cases)*0.25,
        label = "aP", color = "darkred", angle = 90, vjust = -0.5) +
annotate("text", x = 2020, y = max(cdc$cases)*0.25,
        label = "COVID-19", color = "darkblue", angle = 90, vjust = -0.5)
```



What I noticed:

The full introduction of mandatory wP (whole-cell) pertussis immunization in the mid 1940s leads to a dramatic reduction in case numbers (from over 200,000 to 100s).

The switch to the aP (never acellular vaccine) leads to an increasing number of cases, due to the fact that the immunity produced by aP is generally weaker than wP. There may also be other factors contributing to the increasing number of cases.

The 2020 lockdowns and social distancing measures resulted in a decreasing number of cases.

Q3. Describe what happened after the introduction of the aP vaccine? Do you have a possible explanation for the observed trend?

After the introduction of the acellular pertussis (aP) vaccine in 1996, pertussis cases began rising again, with multiple outbreaks in the following decades. One likely explanation is that the aP vaccine produces weaker and more rapidly waning immunity compared to the older whole-cell vaccine, allowing more transmission and susceptibility in adolescents and adults. Additional contributors may include bacteria evolution and improved diagnostic testing using PCR. Social factors, including decreased vaccine uptake due to the modern anti-vaccine movement, may also amplify the resurgence by weakening herd immunity.

The CMI-PB project

The CMI-PB project website: <https://www.cmi-pb.org>. The mission of CMI-PB is to provide the scientific community with a comprehensive, high-quality and freely accessible resource of Pertussis booster vaccination.

They make their data available via JSON format API endpoints – basically the database tables in a key::value type format. To read this, we can use the `read_json()` function from the `jsonlite` package.

```
library (jsonlite)

subject <- read_json(path = "https://www.cmi-pb.org/api/v5_1/subject"
                      , simplifyVector = TRUE)

head (subject)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female Not	Hispanic or Latino	White
2	2	wP	Female Not	Hispanic or Latino	White
3	3	wP	Female	Unknown	White
4	4	wP	Male Not	Hispanic or Latino	Asian
5	5	wP	Male Not	Hispanic or Latino	Asian
6	6	wP	Female Not	Hispanic or Latino	White

	year_of_birth	date_of_boost	dataset
1	1986-01-01	2016-09-12	2020_dataset
2	1968-01-01	2019-01-28	2020_dataset
3	1983-01-01	2016-10-10	2020_dataset
4	1988-01-01	2016-08-29	2020_dataset
5	1991-01-01	2016-08-29	2020_dataset
6	1988-01-01	2016-10-10	2020_dataset

Q. How many individuals are in this dataset?

```
nrow(subject)
```

```
[1] 172
```

Q4. How many aP and wP infancy vaccinated subjects are in the dataset?

```
table(subject$infancy_vac)
```

```
aP wP  
87 85
```

Q5. How many Male and Female subjects/patients are in the dataset?

```
table(subject$biological_sex)
```

```
Female    Male  
    112     60
```

Q. What is the breakdown of race?

```
table(subject$race)
```

```
        American Indian/Alaska Native  
                1  
                Asian  
                44  
        Black or African American  
                5  
        More Than One Race  
                19  
Native Hawaiian or Other Pacific Islander  
                2  
        Unknown or Not Reported  
                21  
                White  
                80
```

Q6. What is the breakdown of race and biological sex (e.g. number of Asian females, White males etc...)?

```
table(subject$race, subject$biological_sex)
```

	Female	Male
American Indian/Alaska Native	0	1
Asian	32	12
Black or African American	2	3
More Than One Race	15	4
Native Hawaiian or Other Pacific Islander	1	1
Unknown or Not Reported	14	7
White	48	32

This breakdown is not particularly representative of the US population.

Investigating the Age Differences

Q7. Using this approach determine (i) the average age of wP individuals, (ii) the average age of aP individuals; and (iii) are they significantly different?

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(lubridate)
```

Attaching package: 'lubridate'

The following objects are masked from 'package:base':

```
date, intersect, setdiff, union
```

```
subject <- subject %>%
  mutate(age = time_length(today() - ymd(year_of_birth), "years"))

subject %>%
  group_by(infancy_vac) %>%
  summarise(avg_age = mean(age, na.rm = TRUE))
```

```
# A tibble: 2 x 2
  infancy_vac avg_age
  <chr>       <dbl>
1 aP         27.8
2 wP         36.6
```

The average age of wP individuals is 36.57 years old. The average age of aP individuals is 27.82 years old.

```
x <- t.test(age ~ infancy_vac, data = subject)
x$p.value
```

```
[1] 2.372101e-23
```

A t-test indicates that the p value is 2.37e-23, far below the common 0.05 cutoff. Therefore, the ages of aP and wP individuals are statistically significantly different.

Q8. Determine the age of all individuals at time of boost?

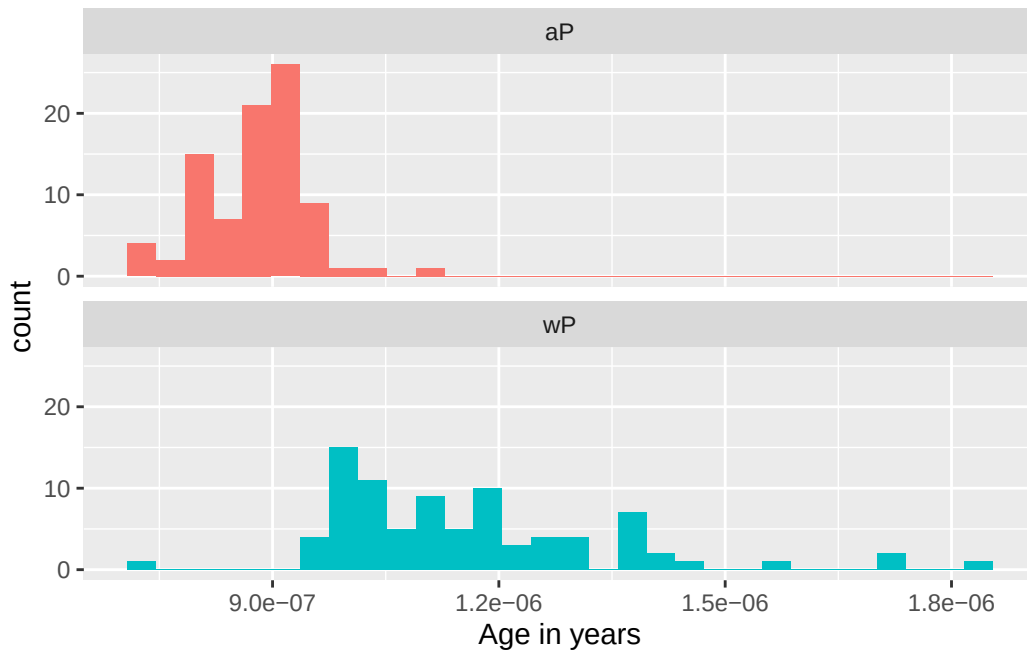
```
int <- ymd(subject$date_of_boost) - ymd(subject$year_of_birth)
age_at_boost <- time_length(int, "year")
head(age_at_boost)
```

```
[1] 30.69678 51.07461 33.77413 28.65982 25.65914 28.77481
```

Q9. With the help of a faceted boxplot or histogram (see below), do you think these two groups are significantly different?


```
ggplot(subject) +
  aes(time_length(age, "year"),
       fill=as.factor(infancy_vac)) +
  geom_histogram(show.legend=FALSE) +
  facet_wrap(vars(infancy_vac), nrow=2) +
  xlab("Age in years")
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



```
t.test(age ~ infancy_vac, data = subject)
```

Welch Two Sample t-test

data: age by infancy_vac

t = -12.918, df = 104.03, p-value < 2.2e-16

alternative hypothesis: true difference in means between group aP and group wP is not equal to 0

95 percent confidence interval:

-10.094058 -7.407351

sample estimates:

mean in group aP mean in group wP

27.81732 36.56802

```
x$p.value
```

```
[1] 2.372101e-23
```

The aP and wP groups differ very strongly in age. The histogram shows that the aP participants cluster tightly around their mid-20s to early 30s, whereas the wP group is substantially older and has a much wider age range extending into the 40s and 50s. A formal Welch two-sample t-test confirms this visual impression, yielding an extremely small p-value ($p = 2.37e-23$), indicating that the age distributions of the two groups are statistically different.

Joining Multiple Tables

```
specimen <- read_json("https://www.cmi-pb.org/api/v5_1/specimen"
                      , simplifyVector = T)
ab_titer <- read_json("https://www.cmi-pb.org/api/v5_1/plasma_ab_titer"
                      , simplifyVector = T)
```

```
head(specimen)
```

	specimen_id	subject_id	actual_day_relative_to_boost	
1	1	1	-3	
2	2	1	1	
3	3	1	3	
4	4	1	7	
5	5	1	11	
6	6	1	32	

	planned_day_relative_to_boost	specimen_type	visit
1	0	Blood	1
2	1	Blood	2
3	3	Blood	3
4	7	Blood	4
5	14	Blood	5
6	30	Blood	6

```
head(ab_titer)
```

	specimen_id	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	1	IgE	FALSE	Total	1110.21154	2.493425
2	1	IgE	FALSE	Total	2708.91616	2.493425
3	1	IgG	TRUE	PT	68.56614	3.736992
4	1	IgG	TRUE	PRN	332.12718	2.602350
5	1	IgG	TRUE	FHA	1887.12263	34.050956
6	1	IgE	TRUE	ACT	0.10000	1.000000

	unit	lower_limit_of_detection
1	UG/ML	2.096133
2	IU/ML	29.170000
3	IU/ML	0.530000
4	IU/ML	6.205949
5	IU/ML	4.679535
6	IU/ML	2.816431

We need to link these tables with the `subject` table, so we can begin to analyze the data and know who a given Ab sample was collected from and when.

Q9. Complete the code to join specimen and subject tables to make a new merged data frame containing all specimen records along with their associated subject details:

```
meta <- inner_join(subject, specimen)
```

Joining with ``by = join_by(subject_id)``

```
dim(meta)
```

```
[1] 1503  14
```

```
head(meta)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female Not Hispanic or Latino	White	
2	1	wP	Female Not Hispanic or Latino	White	
3	1	wP	Female Not Hispanic or Latino	White	
4	1	wP	Female Not Hispanic or Latino	White	
5	1	wP	Female Not Hispanic or Latino	White	
6	1	wP	Female Not Hispanic or Latino	White	

	year_of_birth	date_of_boost	dataset	age	specimen_id
--	---------------	---------------	---------	-----	-------------

1	1986-01-01	2016-09-12	2020_dataset	39.9206	1
2	1986-01-01	2016-09-12	2020_dataset	39.9206	2
3	1986-01-01	2016-09-12	2020_dataset	39.9206	3
4	1986-01-01	2016-09-12	2020_dataset	39.9206	4
5	1986-01-01	2016-09-12	2020_dataset	39.9206	5
6	1986-01-01	2016-09-12	2020_dataset	39.9206	6
	actual_day_relative_to_boost	planned_day_relative_to_boost	specimen_type		
1		-3	0	Blood	
2		1	1	Blood	
3		3	3	Blood	
4		7	7	Blood	
5		11	14	Blood	
6		32	30	Blood	
	visit				
1	1				
2	2				
3	3				
4	4				
5	5				
6	6				

Now, let's also join the `ab_titer` table with the `meta` table.

Q10. Now using the same procedure join meta with titer data so we can further analyze this data in terms of time of visit aP/wP, male/female etc.

```
ab_data <- inner_join(meta, ab_titer)
```

Joining with ``by = join_by(specimen_id)``

```
dim(ab_data)
```

```
[1] 61956    21
```

```
head(ab_data)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female	Not Hispanic or Latino	White
2	1	wP	Female	Not Hispanic or Latino	White
3	1	wP	Female	Not Hispanic or Latino	White

4	1	wP	Female Not Hispanic or Latino White				
5	1	wP	Female Not Hispanic or Latino White				
6	1	wP	Female Not Hispanic or Latino White				
	year_of_birth	date_of_boost	dataset	age	specimen_id		
1	1986-01-01	2016-09-12	2020_dataset	39.9206	1		
2	1986-01-01	2016-09-12	2020_dataset	39.9206	1		
3	1986-01-01	2016-09-12	2020_dataset	39.9206	1		
4	1986-01-01	2016-09-12	2020_dataset	39.9206	1		
5	1986-01-01	2016-09-12	2020_dataset	39.9206	1		
6	1986-01-01	2016-09-12	2020_dataset	39.9206	1		
	actual_day_relative_to_boost	planned_day_relative_to_boost	specimen_type				
1		-3	0	Blood			
2		-3	0	Blood			
3		-3	0	Blood			
4		-3	0	Blood			
5		-3	0	Blood			
6		-3	0	Blood			
	visit	isotype	is_antigen_specific	antigen	MFI	MFI_normalised	unit
1	1	IgE	FALSE	Total	1110.21154	2.493425	UG/ML
2	1	IgE	FALSE	Total	2708.91616	2.493425	IU/ML
3	1	IgG	TRUE	PT	68.56614	3.736992	IU/ML
4	1	IgG	TRUE	PRN	332.12718	2.602350	IU/ML
5	1	IgG	TRUE	FHA	1887.12263	34.050956	IU/ML
6	1	IgE	TRUE	ACT	0.10000	1.000000	IU/ML
	lower_limit_of_detection						
1		2.096133					
2		29.170000					
3		0.530000					
4		6.205949					
5		4.679535					
6		2.816431					

Q. How many Ab measurements do we have in total?

```
nrow(ab_data)
```

```
[1] 61956
```

Q. How many different isotypes (types of Ab) are in the dataset?

```
unique(ab_data$isotype)
```

```
[1] "IgE" "IgG" "IgG1" "IgG2" "IgG3" "IgG4"
```

Q. How many different antigens?

```
unique(ab_data$antigen)
```

```
[1] "Total" "PT" "PRN" "FHA" "ACT" "LOS" "FELD1"  
[8] "BETV1" "LOLP1" "Measles" "PTM" "FIM2/3" "TT" "DT"  
[15] "OVA" "PD1"
```

Q11. How many specimens (i.e. entries in abdata) do we have for each isotype?

```
table(ab_data$isotype)
```

```
 IgE  IgG  IgG1  IgG2  IgG3  IgG4  
6698 7265 11993 12000 12000 12000
```

Q12. What are the different \$dataset values in ab_data and what do you notice about the number of rows for the most “recent” dataset?

```
table(ab_data$dataset)
```

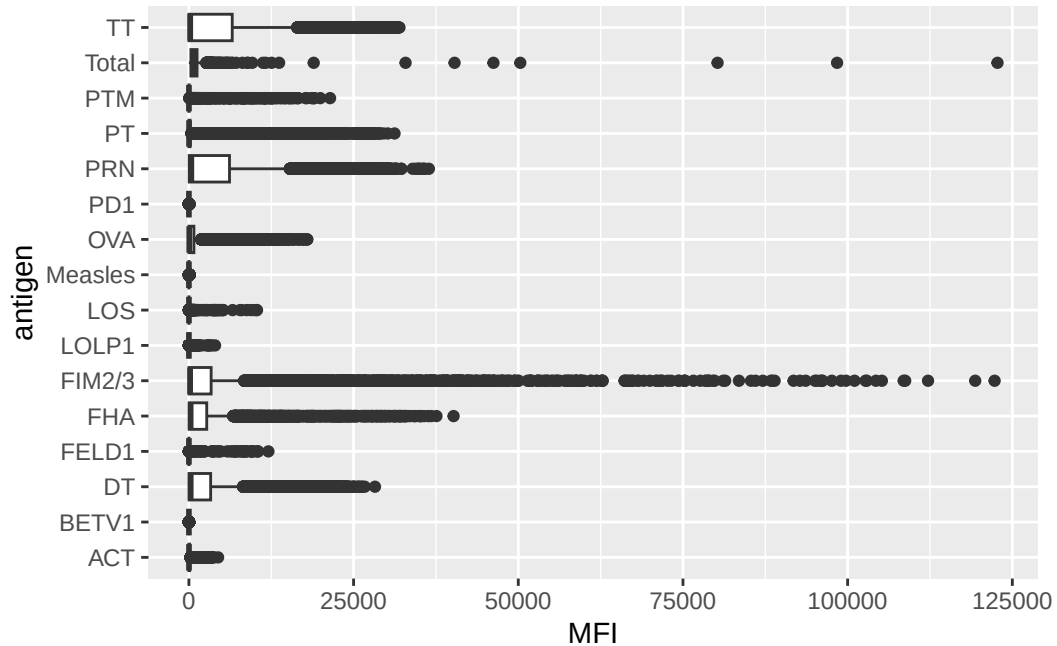
```
2020_dataset 2021_dataset 2022_dataset 2023_dataset  
      31520           8085           7301           15050
```

The dataset column contains four values: 2020_dataset, 2021_dataset, 2022_dataset, and 2023_dataset. The number of rows differs greatly across years: 2020 has by far the largest number of records (31,520), while 2022 and 2021 are much smaller (~7–8k). Interestingly, the most “recent” dataset (2023) does not have the largest number of rows – it has only 15,050 entries, about half of 2020. This suggests that more recent data collection has been reduced, is still incomplete or reflects changes in sampling effort or assay prioritization.

Examining IgG Ab Titer levels

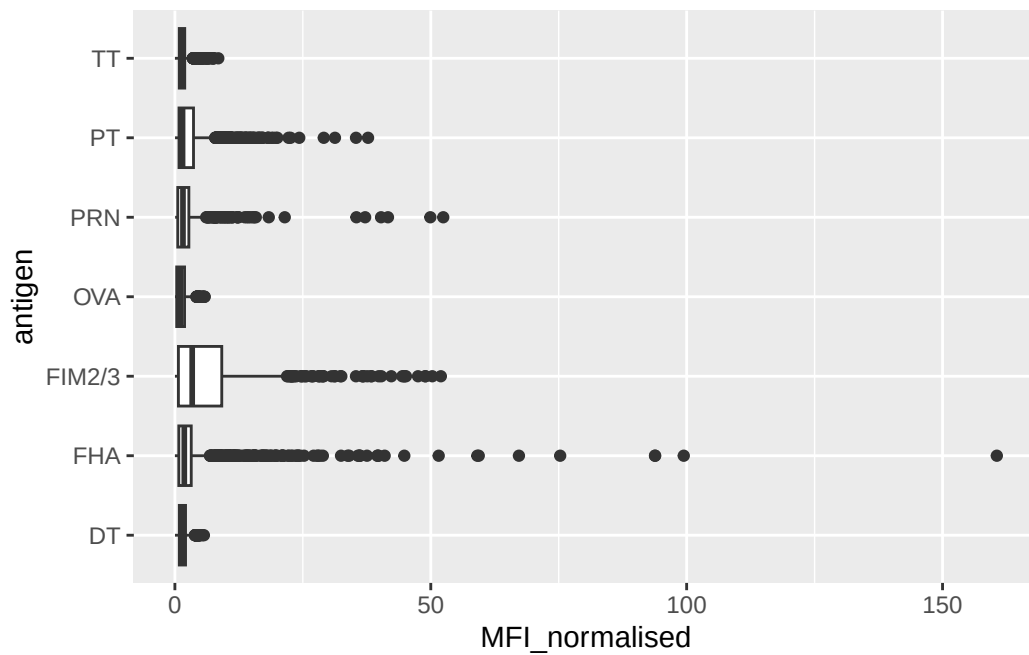
```
ggplot(ab_data) +
  aes(MFI, antigen) +
  geom_boxplot()
```

Warning: Removed 1 row containing non-finite outside the scale range (`stat\_boxplot()`).



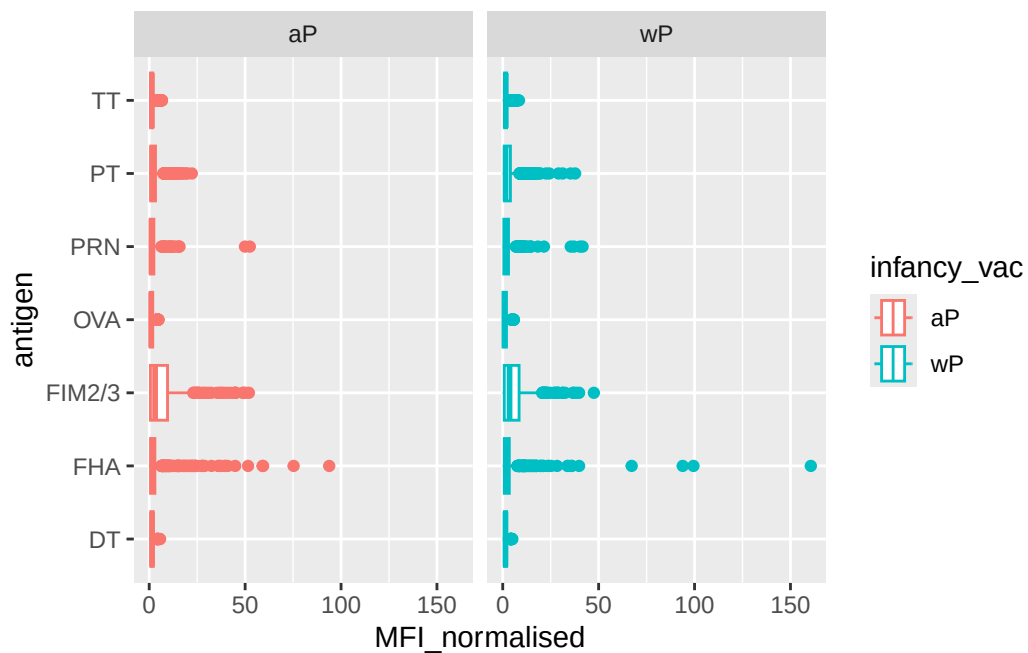
```
igg <- ab_data |>
  filter(isotype == "IgG")
```

```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot()
```



We can “facet” our plot by wP vs aP

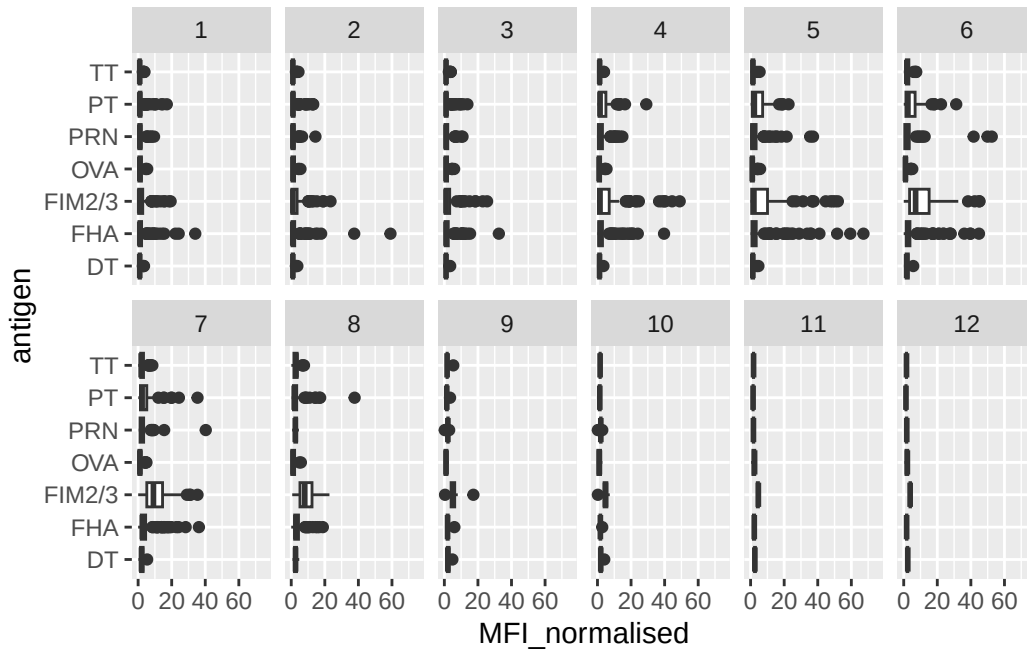
```
ggplot(igg) +
  aes(MFI_normalised, antigen, col = infancy_vac) +
  geom_boxplot() +
  facet_wrap(~infancy_vac)
```

Q13. Complete the following code to make a summary boxplot of Ab titer levels (MFI) for all antigens:

```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot() +
  xlim(0,75) +
  facet_wrap(vars(visit), nrow=2)
```

Warning: Removed 5 rows containing non-finite outside the scale range (`stat\_boxplot()`).



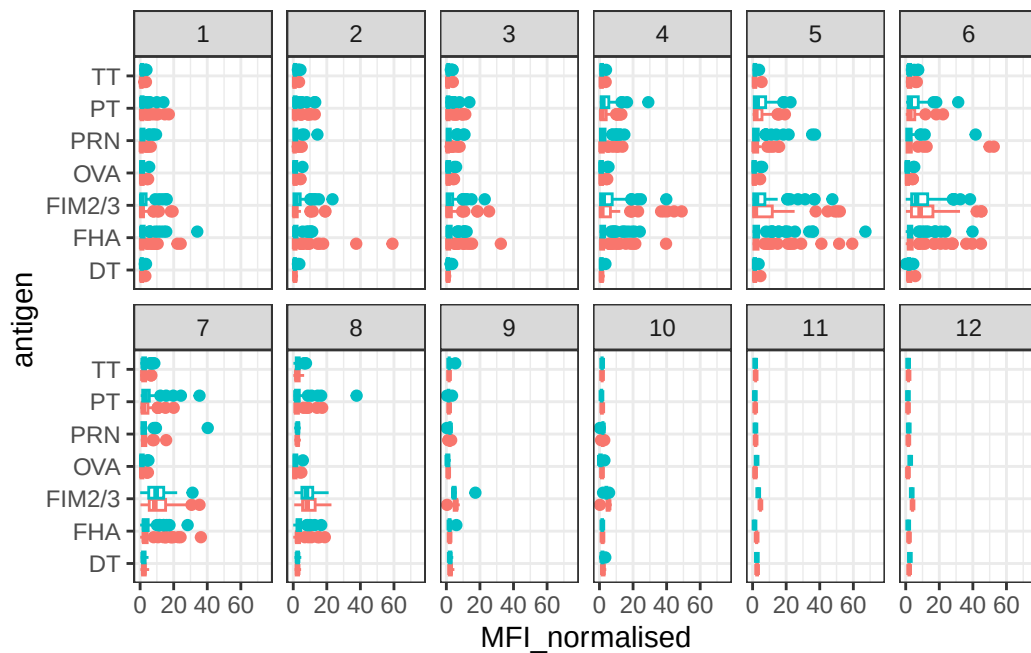
Q14. What antigens show differences in the level of IgG antibody titers recognizing them over time? Why these and not others?

Several pertussis-related antigens show clear changes in IgG levels over time, particularly PT, PRN, FHA and FIM2/3. These antigens rise noticeably after the early visits (around visits 4–7), showing boosted antibody responses following vaccination, and then gradually decline, reflecting antibody waning. In contrast, antigens like TT, DT and the negative control OVA remain relatively constant across all visits, showing no meaningful time-dependent change. This pattern occurs because only the pertussis antigens (PT, PRN, FHA and FIM2/3) are present in the Tdap booster dose administered during the study, whereas TT/DT were not boosted and OVA is unrelated to vaccination. As a result, only pertussis-specific antigens exhibit dynamic IgG responses over the 12-visit timeline.

We can also add some colors to the plots.

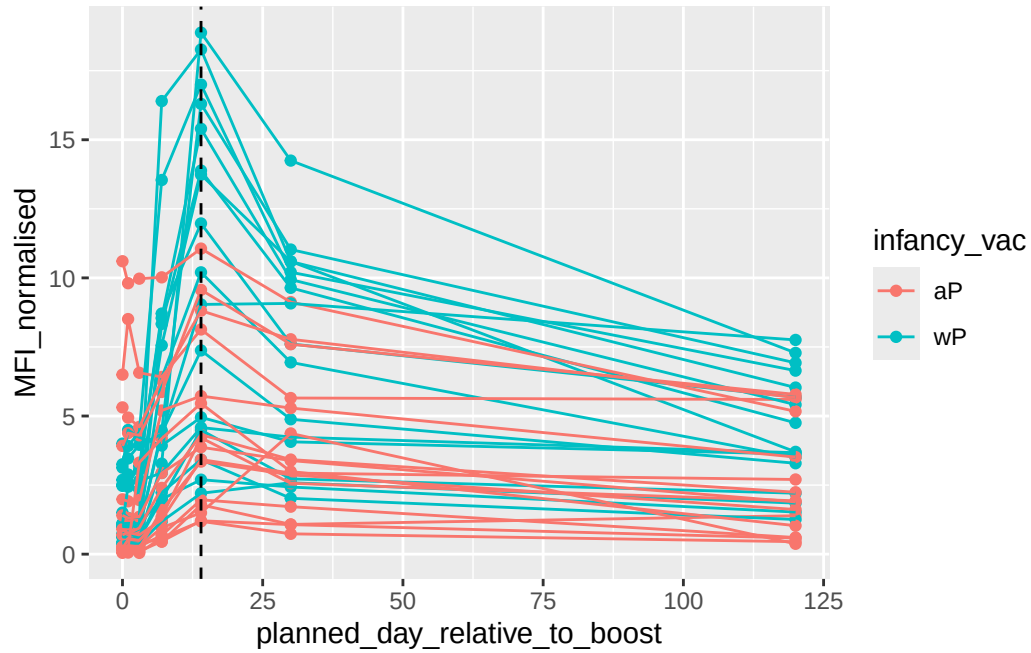
```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac ) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit), nrow=2) +
  xlim(0,75) +
  theme_bw()
```

Warning: Removed 5 rows containing non-finite outside the scale range (``stat_boxplot()``).



More advanced analysis digging into individual antigen responses over time

```
filter(igg, antigen == "PT", dataset == "2021_dataset") |>
ggplot() +
  aes(x=planned_day_relative_to_boost,
      y=MFI_normalised,
      col=infancy_vac,
      group=subject_id) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept=14, linetype = "dashed")
```



This plot compares PT-specific IgG responses after boosting between individuals primed in infancy with either whole-cell (wP) or acellular (aP) pertussis vaccines. Antibody levels rise rapidly and peak at 14 days post-boost (vertical dashed line), then gradually decline. Across the entire time course, wP-primed individuals show higher peak responses and sustain higher antibody levels than aP-primed individuals, demonstrating the more robust and durable recall immunity associated with wP priming.